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Case Study

On

TripMate – Travel Planning Website

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1. Introduction

TripMate is a front-end travel planning website developed using HTML, CSS and JavaScript. The main purpose of the application is to provide users with a simple platform to explore travel destinations, select a destination and prepare a basic trip plan.

The website displays ten selected destinations, including Manali, Shimla, Kasol, Dharamshala, Dalhousie, Spiti Valley, Srinagar, Gulmarg, Pahalgam and Leh. Each destination contains information such as its location, description and estimated starting price.

Users can search for destinations, apply filters and select a destination for trip planning. The booking planner allows users to enter travel details such as departure date, trip duration, transportation mode, number of passengers and travel class.

JavaScript processes these selections and calculates an estimated trip cost. Users can add their planned trips to a cart, review the details, remove trips and view the combined estimated amount.

The project demonstrates how HTML, CSS and JavaScript can be combined to create an interactive and user-friendly web application. It is an educational front-end project and does not provide actual travel reservations or payment processing.

2. Problem Statement

Planning a trip involves several decisions, including selecting a destination, choosing travel dates, deciding the mode of transportation and estimating the overall expenses.

Managing these details separately can make the planning process confusing and time-consuming. Users may also find it difficult to compare different destinations and understand how their travel preferences affect the estimated cost.

TripMate addresses this problem by bringing basic travel exploration and planning features together in one website. It allows users to browse destinations, select travel preferences, calculate an estimated trip cost and manage planned trips through a cart system.

3. Objectives

The main objectives of the TripMate project are:

- To develop a simple and user-friendly travel planning website.
- To create the website using HTML, CSS and JavaScript.
- To display ten selected travel destinations with their descriptions and estimated prices.
- To implement destination searching and filtering functionality.
- To allow users to select destinations and enter their travel preferences.
- To provide different transportation options such as Flight, Train, Bus and Car.
- To calculate an estimated trip cost based on user selections.
- To implement a cart system for managing planned trips.
- To understand DOM manipulation, event handling and form validation.
- To apply practical web development concepts through a real-world-inspired project.

4. Software and Technologies Used

- **HTML5:** Used to create the structure of the website, including navigation, destination sections, booking forms and cart elements.
- **CSS3:** Used to design the website layout, colours, typography, destination cards, buttons, forms and responsive appearance.
- **JavaScript:** Used for dynamic destination display, search and filtering, destination selection, price calculation and cart management.
- **Visual Studio Code:** Used as the code editor for developing and managing the project files.
- **Web Browser:** Microsoft Edge or Google Chrome is used to run and test the website.
- **External Image Resources:** Image URLs are used to display travel destination and background images.

The project uses vanilla JavaScript without requiring a separate JavaScript framework or backend server.

5. Destination Data Description

The project uses a predefined JavaScript array to store destination information. Each destination contains details such as its name, region, state, estimated base price, theme, description and image reference.

The website currently contains ten destinations.

Destination	State / Region	Base Price
Manali	Himachal Pradesh	₹5,999
Shimla	Himachal Pradesh	₹4,999
Kasol	Himachal Pradesh	₹3,999
Dharamshala	Himachal Pradesh	₹4,499
Dalhousie	Himachal Pradesh	₹5,499
Spiti Valley	Himachal Pradesh	₹8,999
Srinagar	Jammu and Kashmir	₹9,999
Gulmarg	Jammu and Kashmir	₹10,999
Pahalgam	Jammu and Kashmir	₹8,999
Leh	Ladakh	₹12,999

The main data handled by the system is described below:

Data	Purpose
Destination details	Stores destination name, location, description and base price.
Search input	Accepts the destination name or location entered by the user.
Filter selection	Filters destinations according to the selected region or state.
Booking details	Stores travel date, duration, transport and passenger information.
Price estimate	Displays the calculated estimated trip cost.
Cart data	Holds the trips added by the user during the current session.

The destination information is predefined in the JavaScript code. The cart information is maintained in application memory and is not permanently stored in a database.

6. Methodology and Workflow

Requirement Analysis: The project requirements were identified by considering the basic steps involved in planning a trip, including destination selection, travel preferences and cost estimation.

Website Structure: HTML was used to create the navigation bar, home section, destination section, selected destination area, booking planner, cart and footer.

User Interface Design: CSS was used to style the website and create destination cards, forms, buttons, layouts and responsive elements.

Destination Management: Destination information was stored in a JavaScript array. JavaScript uses this data to generate destination cards and populate the destination selection options.

Search and Filtering: The search input and filter controls allow users to find destinations according to their names and locations.

Destination Selection: When a user selects a destination, the website updates the selected destination section and the booking planner.

Trip Planning: The user enters the departure date, trip duration, transportation option, number of travellers and travel class.

Price Calculation: JavaScript reads the selected values and calculates an estimated trip cost using predefined multipliers.

Cart Management: The selected trip is added to the cart, where the user can review the details, remove trips and view the combined estimated price.

Testing: The website can be checked by selecting different destinations, searching and filtering results, changing booking inputs, adding trips to the cart and checking the estimated totals.

7. Main System Features

- **Destination Display:** Displays ten selected travel destinations with their images, descriptions and estimated starting prices.
- **Search Functionality:** Allows users to search for destinations by name or location.
- **Destination Filtering:** Allows users to filter destinations according to the available region or state options.
- **Destination Selection:** Displays the selected destination's information and updates the booking planner.
- **Trip Planner:** Accepts departure date, trip duration, transportation mode, passenger count and travel class.
- **Transportation Options:** Provides Flight, Train, Bus and Car as transportation choices.
- **Dynamic Price Estimation:** Calculates the estimated trip cost according to the selected travel preferences.
- **Cart Management:** Allows users to add trips, view planned trips and remove unwanted entries.
- **Total Calculation:** Displays the combined estimated price of the trips added to the cart.
- **Demo Confirmation:** Provides a demonstration booking confirmation without processing an actual payment.

8. SOURCE CODE – IMPORTANT LOGICAL PARTS

The main functionality of TripMate is implemented using JavaScript. Instead of including the complete source code, the following snippets demonstrate the important logic used in the project.

8.1 Destination Data

A JavaScript array stores destination details such as name, state and base price.

```
const destinations = [
    { name: "Manali", state: "Himachal Pradesh", price: 5999 },
    { name: "Shimla", state: "Himachal Pradesh", price: 4999 },
    { name: "Kasol", state: "Himachal Pradesh", price: 3999 },
    { name: "Dharamshala", state: "Himachal Pradesh", price: 4499 },
    { name: "Dalhousie", state: "Himachal Pradesh", price: 5499 },
    { name: "Spiti Valley", state: "Himachal Pradesh", price: 8999 },
    { name: "Srinagar", state: "Jammu and Kashmir", price: 9999 },
    { name: "Gulmarg", state: "Jammu and Kashmir", price: 10999 },
    { name: "Pahalgam", state: "Jammu and Kashmir", price: 8999 },
    { name: "Leh", state: "Ladakh", price: 12999 }
];
```

8.2 Destination Search and Filtering

This logic filters destinations according to the search text and selected region.

```
function filterDestinations(search, region) {
    return destinations.filter(destination => {
        const matchesSearch =
            destination.name.toLowerCase().includes(search.toLowerCase()) ||
            destination.state.toLowerCase().includes(search.toLowerCase());

        const matchesRegion =
```

```
            region === "All" || destination.region === region;

        return matchesSearch && matchesRegion;
    });
}
```

8.3 Trip Price Calculation

JavaScript calculates the estimated trip cost using the destination price, number of passengers, transportation and travel class.

```
function calculatePrice(basePrice, travellers, transport, travelClass, days) {
    const transportRates = {
        Flight: 1.35,
        Train: 1.00,
        Bus: 0.80,
        Car: 1.20
    };

    const classRates = {
        Standard: 1.00,
        Premium: 1.20
    };

    const durationRate = 1 + Math.max(0, days - 3) * 0.12;

    return Math.round(
        basePrice * travellers *
        transportRates[transport] *
        classRates[travelClass] *
        durationRate
    );
}
```

8.4 Cart Management

The cart stores planned trips, allows users to remove entries and calculates the combined estimated price.

```
let cart = [];

function addToCart(trip) {
    cart.push(trip);
    displayCart();
}

function removeFromCart(index) {
    cart.splice(index, 1);
    displayCart();
}

function calculateCartTotal() {
    return cart.reduce((total, trip) => total + trip.price, 0);
}
```

8.5 Dynamic Website Updates

JavaScript uses DOM manipulation and event handling to update destination information, booking details and cart information according to user actions.

```
document.addEventListener("DOMContentLoaded", () => {
    renderDestinations();
    initialise();
});
```

9. Reported Results

The TripMate website provides a single interface for browsing destinations and preparing basic travel plans.

The main results demonstrated by the application are:

Test Scenario	Expected Result
Open the website	Home page and website sections are displayed.
View destinations	Ten destination cards are available.
Search for Manali	Matching destination results are displayed.
Apply a state filter	Destinations matching the selected filter are displayed.
Select a destination	The selected destination information is updated.
Change booking details	The estimated trip price is recalculated.
Add a planned trip	The selected trip appears in the cart.
Remove a planned trip	The selected cart entry is removed and the total is updated.
Confirm a demo booking	A demonstration confirmation is displayed.

The project demonstrates the practical use of HTML, CSS and JavaScript to create a dynamic travel planning interface. The estimated prices are based on predefined values and calculation rules rather than live travel provider information.

10. Limitations

- The destination list is predefined and cannot be managed through an administrator panel.
- The displayed destination prices are sample estimates and are not connected to live travel providers.
- The website does not provide actual flight, train, bus or car reservations.
- The cart information is stored in application memory and is lost when the page is refreshed.
- The project does not include user registration or login functionality.
- There is no database or backend system for permanent storage.
- The demo confirmation does not process payments or create actual bookings.
- External images may require an internet connection to load.

11. Conclusion

TripMate demonstrates how a basic travel planning website can be developed using HTML, CSS and JavaScript.

The project brings together destination exploration, search and filtering, trip planning, price estimation and cart management in a single interface. It provides users with a simple way to select destinations and understand estimated travel expenses according to their preferences.

Through this project, practical knowledge of HTML page structure, CSS styling, JavaScript programming, DOM manipulation, event handling and form validation has been applied.

Although TripMate is a front-end educational application, it provides a foundation that can be extended into a more advanced travel planning and booking platform.

12. Learning Outcomes

- Learned how to structure a website using HTML5.
- Understood how CSS3 is used to design and style web pages.
- Practised JavaScript programming for interactive website functionality.
- Learned how to store and manage information using JavaScript arrays and objects.
- Understood dynamic HTML generation and DOM manipulation.
- Implemented destination searching and filtering.
- Learned how to handle form inputs and user events.
- Applied mathematical calculations to estimate trip costs.
- Understood how to manage cart entries using JavaScript.
- Gained practical experience in organising and testing a front-end web application.